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Transmission and Reception of Longitudinally-Polarized Momentum Waves

This introduction to vector potential radiation describes the verification of an alternative form of radiation predicted by James Clerk Maxwell's famous equations of electromagnetic radiation.

The world has been communicating with transverse electromagnetic (**TEM**) radio waves for more than one-hundred years. These are the waves known to radio amateurs, with the electric field, **E,** perpendicular to the magnetic field, **H**, and both fields perpendicular to the direction of propagation.

Research at McMaster University in Hamilton, Ontario has recently documented a new type of radiation, called vector potential (VP) radiation, which shares some similarities with TEM radiation, but is distinct in many ways:

1. Both TEM radio waves and VP waves were predicted by James Clerk Maxwell.
2. Both waves travel at the speed of light.
3. Both waves are emitted by a current-carrying wire.
4. Both wave amplitudes diminish with range as 1/Range.

There the similarities end:

1. VP waves are longitudinal waves, much like sound waves.
2. VP waves carry no energy (The ultimate QRP). VP waves are carried by "virtual photons" rather than the real photons of radio waves.
3. VP waves carry only linear momentum. In fact, Maxwell's term for VP waves was "electro-kinetic momentum."
4. VP waves cannot be received with metallic antennas. For this reason, they have remained a mystery since Maxwell described them.
5. Since VP waves cannot be received with normal metallic antennas, they are in a fundamental way covert in nature. It is expected that one of the first uses of VP communication will be for the military, in as much as transmissions cannot be easily intercepted.

In the years ahead, radio amateurs may play an important role in learning more about VP radiation. This article may serve as a primer.

Basic Characteristics

VP waves are emitted by all current carrying wires, in direct proportion with the current amplitude along the wire, and the length of the wire in wavelengths. In fact, a normal half-wave dipole emits VP waves along with the usual TEM radio waves. The vector potential (called **A** in graduate electromagnetic texts), however, *is always polarized parallel with the current carrying wire*. This stands at odds with **E** and **H** fields, which tend to fold around a radiating wire. Most importantly, the vector potential exists *even off the ends of a dipole*, where there is no radiated **E** or **H** field. This makes the region off the ends of a dipole of special interest, as **A** exists in the absence of **E** or **H** fields. In this region, the vector potential is polarized entirely longitudinally; in other words, in the direction of propagation.

These VP waves carry mechanical momentum. Free electrons oscillate remarkably well when they are exposed to a VP wave. How can we convert the VP wave back into an RF signal, though? Fortunately, this problem has now been solved: The solution is to use a fluorescent tube (with electronic "plasma" inside) which is folded in the shape of the letter U. These tubes are usually sold at the hardware store for about $6.00, for use in camping lanterns. Figure 1 schematically shows such a U tube aligned with a VP wave coming from the left side of the drawing. The VP wave travels the length of the tube, and causes an RF current to travel down the tube structure, similar to a parallel transmission line. When the RF current reaches the tube terminals (at the end of the tube), this current continues onward into a coaxial cable attached to the terminals and on to a radio receiver.

Momentum Coupling

The **A** vector potential will couple linear momentum to free-electrons in its path. This is the basis for the detection of VP waves, and is called the Aharonov-Bohm Effect. (To learn more about the Aharonov-Bohm Effect, do a Google search on the term.) Electrons in copper and other metals can never move at velocities greater than about 3 mm/s because of crystalline lattice scattering, which becomes dominant at that velocity. This is entirely too slow to couple to vector potential radiation.

Free electrons in plasmas can easily travel at "drift velocities" of tens of kilometers per second. This makes plasma antennas a good match for vector potential wave reception. As an example, the vector potential near a half-wave dipole carrying a 10 mA rms current is about 10^{-10} webers/m (SI units). In a well designed plasma detector tube, momentum coupling for this potential (if the tube is aligned with **A**) will yield a free electron velocity of ±20 m/s. Obviously, the sluggish electrons in copper cannot be expected to couple to this — metallic receiving antennas are out.

Figure 1 demonstrates the desired geometry for vector potential coupling momentum to a U-shaped plasma tube, and describes how RF current is developed in the tube terminals.

A Transmitting Antenna for 1296 MHz

This section describes the construction of a VP transmitting antenna for 1296 MHz. Since a transmitting dipole in free space will emit a large TEM RF signal, we house the transmitting element inside a circular waveguide to prevent radiation of TEM power. This shielding stops emission of TEM RF power while allowing the transmission of VP radiation through an open-ended guide. The circular waveguide shown in Figure 2 operates in the transverse magnetic TM_{01} longitudinal mode. The fields inside such circular guide are shown in Figure 3. The cutoff wavelength for this mode is 2.613 times the radius of the guide. Accordingly, a 7 inch diameter galvanized stovepipe (diameter = 17.78 cm) provides a cutoff wavelength of $\lambda_{cut} = (17.78/2) \times 2.613 = 23.23$ cm, which is a frequency of 1291 MHz.

To get maximum current on the waveguide probe (and hence maximum radiated vector potential amplitude) from a given transmitter, the probe should provide an impedance of zero ohms at the coaxial drive port. (Remember, we do not wish to radiate RF power anyway.) To get zero ohms at the drive port, with the other end of the guide as an "open circuit" for the TM_{01} mode, requires that the guide be a quarter wavelength overall from the open port to the coaxial port. For this 7 inch guide, the guide wavelength at 1296 MHz ($\lambda_0 = 23.148$ cm) is:

$$\lambda_{guide} = \frac{\lambda_o}{\sqrt{1-\left(\frac{\lambda_o}{\lambda_{cut}}\right)^2}} = 275 \text{ cm}$$

Accordingly, the waveguide should be 69 cm in length to yield a quarter-wavelength. The coaxial probe should also be this length. It is important to keep the probe longitudinally centered in the waveguide. For this a Styrofoam strut may be used inside the guide.

Zero ohms will yield an infinite SWR on the coaxial cable leading to the transmitter. To prevent transmitter damage, use a 3 dB attenuator pad on the coaxial connection to the waveguide. (Of course, a circulator may also be used with a load on the third port.) A 3 dB pad will provide a return loss of 6 dB for an SWR of 3, which should be adequate protection for most Amateur Radio transmitters. Be sure to use a pad rated at the full transmitter power.

To prevent minor off-axis TEM radiation, a "choke collar" may be used at the open end of the guide. The length of the choke should be 23.148 / 4 = 5.8 cm. This addition, although not strictly essential, will prevent currents on the inside of the waveguide from "folding around" the open end and radiating.

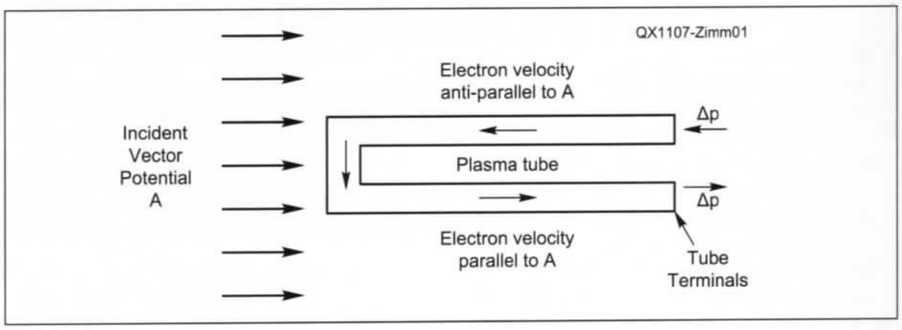

Figure 1 — In this diagram, the vector potential wave is traveling to the right towards the plasma tube (shown at an instant in time). The polarity of A is constantly oscillating at the frequency of the transmitting antenna, *f*. The electron dc current in the tube, provided by the tube high-voltage bias supply, travels in the direction shown by the arrows in the tube itself. At the instant shown, A is anti-parallel with the electron velocity in the top leg of the tube. This yields a momentum increment (and acceleration) to the left as shown. Likewise, in the bottom leg of the tube, A is parallel with the electron velocity, yielding momentum (and acceleration) to the right as shown. (Keep in mind that the momentum contributed to the electrons will be oscillating at frequency *f*.) This changing electron momentum and velocity is the detected RF current in the tube terminals.

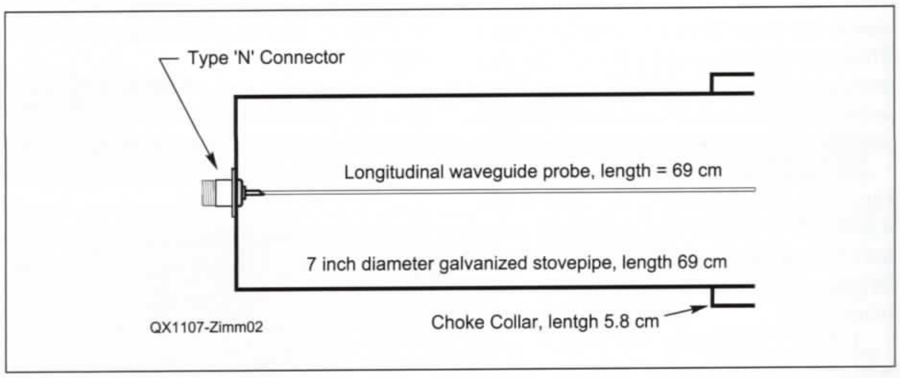

Figure 2 — This drawing shows the transmitting waveguide antenna for VP radiation at 1296 MHz.

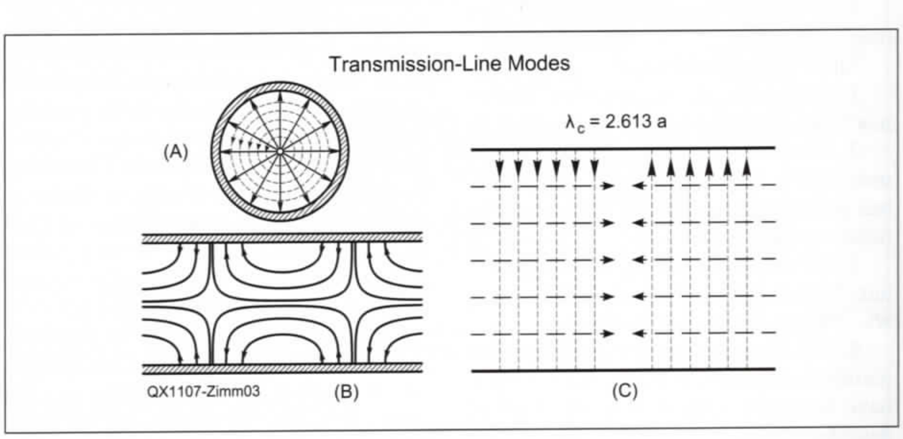

Figure 3 — The longitudinal fields inside a TM_{01} mode circular waveguide are illustrated here. The drawing is from N. Marcuvitz, *Waveguide Handbook*, MIT Radiation Laboratory Series, Vol 10, pg 68, McGraw-Hill, 1951.

A Receiving Antenna for 1296 MHz

The receiving plasma detector is made from a compact mercury-vapor fluorescent tube as found in camping lanterns. The fluorescent lamp is mounted inside the 7 inch diameter waveguide, as shown in Figure 4. At McMaster University, we use commercial waveguide with flanges; the fluorescent tube mounted on its flange is shown in Figure 5.

The U-shaped folded lamp, when ionized with direct current, provides a parallel transmission line with a characteristic impedance of 50 Ω; a perfect match to the 50 Ω coaxial cable. When the longitudinal VP wave enters the waveguide, it couples a travelling wave onto the tube due to the Aharonov-Bohm effect. One leg of the tube is connected to the coaxial center conductor, while the remaining leg is grounded. (A balun could be used here, but the benefits are minor.)

The tube is operated at 200 mA dc current from a high voltage current-regulated power supply coupled to the coaxial line with a bias T. My power supply would not hold the current constant until I put a 10 W 120 Ω resistor in series with the power supply. (Without the resistor, the tube flashed at a 1 Hz rate.) It is important to use a linear supply to keep bias current noise to a minimum.

The negative lead of the power supply should be connected to the coaxial center lead. About –400 V dc is momentarily required to ionize the fluorescent tube. My power supply is limited to –200 V dc. Accordingly, I went to Home Depot and purchased a battery operated fluorescent lamp for use in a closet. I wired the transformer output of this battery operated supply in series with the main power supply. By momentarily energizing the battery operated supply with a push button, the fluorescent tube will ionize, and then the direct current is regulated by the main supply and the 120 Ω resistor. Note that the battery operated supply produces 500 V ac at 50 kHz. Be sure your bias T can momentarily support this large voltage or you will burn out your receiver front end.

Bias T Design

A suitable Bias T for use here is very costly if purchased commercially. Most of these bias Ts will not withstand the momentary 500 V used to ignite the fluorescent tube. This means you probably have to build your own. Figure 6 shows the location of the bias T in the receiver setup. Figure 7 presents a design based on a quarter-wave stub at 1296 MHz. In free-space, a quarter wavelength is 5.8 cm at this frequency. If built on microstrip with Teflon substrate (dielectric constant = 2.1) the stub should be 4.0 cm in length.

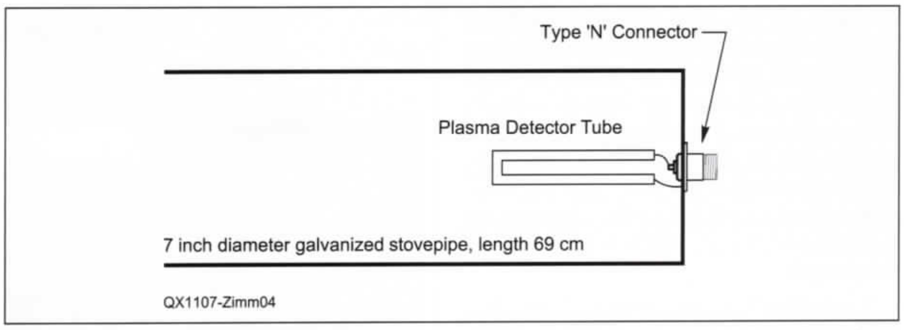

Figure 4 — The receiving waveguide antenna for VP radiation at 1296 MHz uses a camping lantern fluorescent tube. DC bias current to the plasma tube is provided by way of the coaxial cable via an inline bias T network.

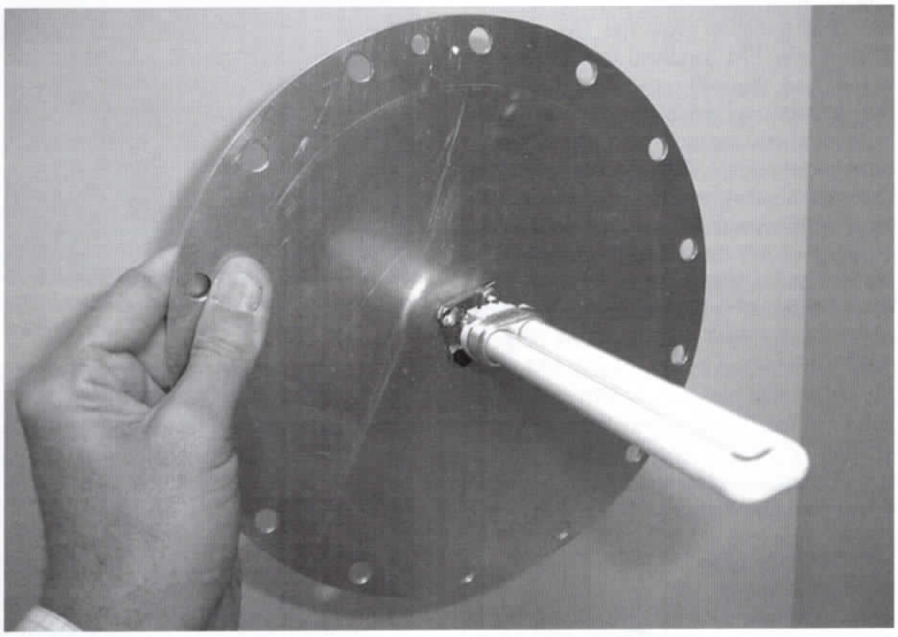

Figure 5 — Here is a photo of the fluorescent plasma detector tube mounted on a waveguide flange.

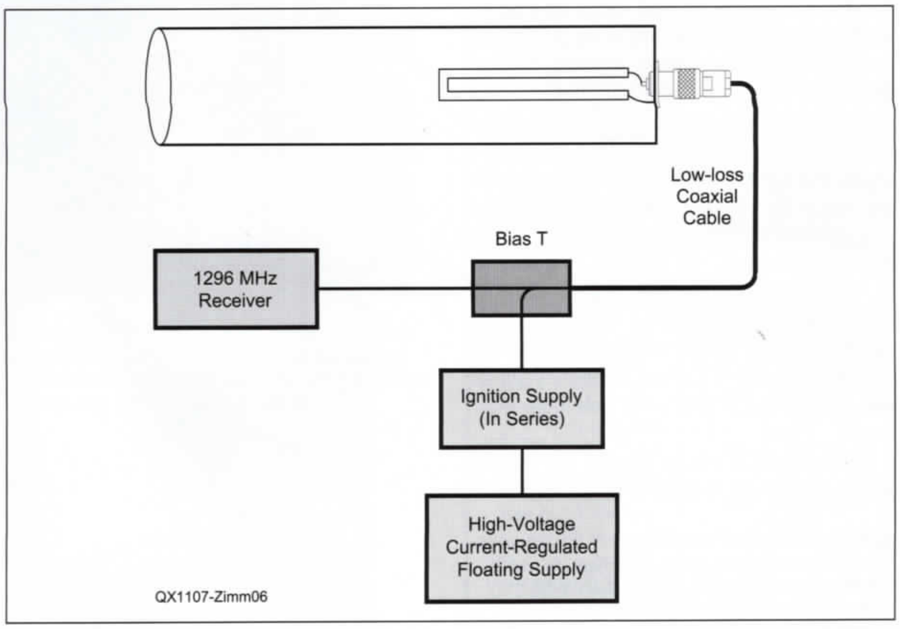

Figure 6 — Block diagram of the receiving setup. The bias T combines RF and tube bias on the single coaxial line leading to the waveguide antenna.

There are two 1000 pF capacitors in the design: one to block dc from the receiver line, and one to bypass the "cold-end" of the stub. In order to withstand the high dc voltage, these capacitors should have Teflon dielectric; for best RF performance, the capacitors should be single layer. If 0.010 inch thick Teflon film is used, a 1 square cm capacitor will yield > 730 pF, which is sufficient for the design[1].

Operating

The open-ended waveguides have an on-axis gain for VP radiation of about 3 dBi. (Additionally, there is an on-axis null for TEM radiation from the TM_{01} guide.) With a 1 W FM handheld transmitter, and a 3 dB pad, this will provide for a range of 500 meters over a clear path. For further range, you may use more drive power to yield more probe current. Alternatively, if you like sheet-metal work as I do, you can build a horn antenna for the transmitter and receiver. Figure 8 shows the 20 dBi horn I built for the receiver, which permitted a solid link at 1500 meters. If you use such horns at both the transmitter and receiver, a link range of 10 kilometers should be realized. FM is not the only mode possible — SSB or CW are certainly fine as well. Figure 9 shows Sherry Goeller, VE3DCU, operating the transmitting station on the 1500 meter link. Figure 10 is a photo of the author at the receiving site.

Summary

Fundamental advances in electromagnetics have yielded a new means of communication that small-signal microwave hams will find of interest; these waves are electromechanical in nature and carry no energy. Perhaps someday radio and television broadcasts will be transmitted without energy! Note that vector potential radiation requires radio frequency current into zero ohms, rather than RF power itself. This means that an intense VP wave may be transmitted without much (or any) RF power. Maybe distant civilizations are communicating with such waves right now?

There are no international regulations governing transmission of VP momentum radiation. It is prudent to operate within Amateur Radio allocated frequency bands and identify your station by call sign, however.

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[1]McMaster-Carr industrial supply house sells 0.010 inch Teflon sheet. Digi-Key Inc. sells adhesive copper foil. These materials may be used to construct the two capacitors. Use no more than a 25 W soldering iron when connecting the capacitors in the bias T circuit.

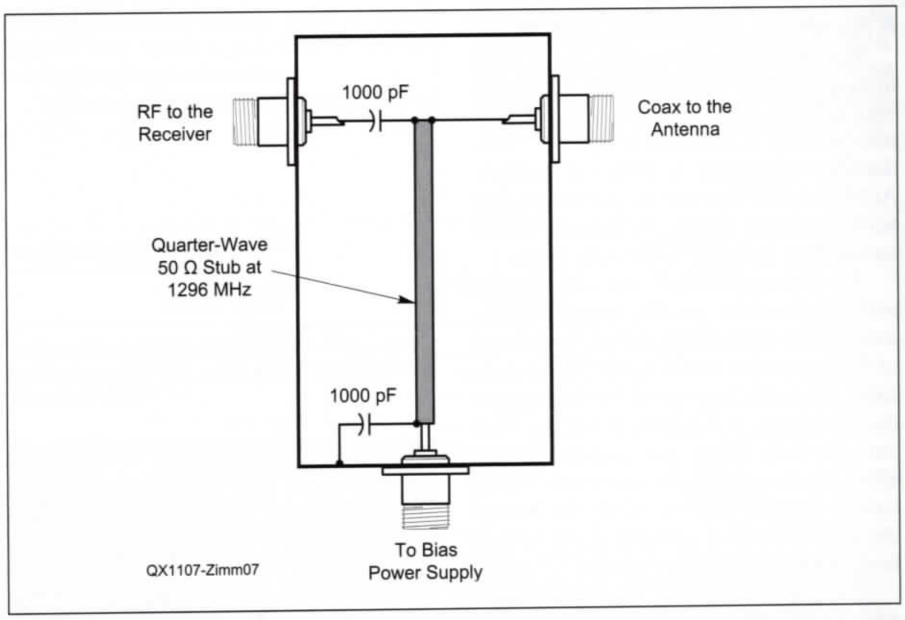

Figure 7 — Design of the bias T used to feed bias current to the waveguide antenna, and route RF back to the receiver from the antenna. The stub may be realized with microstrip. Alternatively, and perhaps easier, is to use a segment of Teflon-dielectric subminiature coaxial cable. If this is done, be sure to intimately ground the coaxial shield at both ends of the stub. Be careful to mark the ports clearly, for if the bias T is reversed your receiver will be destroyed.

Figure 8 — The 20 dBi gain conical horn built for the receiving detector is shown here.

Figure 9 — Sherry Goeller, VE3DCU, operating the transmitting momentum station on the 1500 m link. Note that the coaxial cable is connected to the rear of the waveguide.

Figure 10 — The author, NP4B/VE3RKZ, on the receiving end of the 1500 m momentum link. Note that the receiving horn had to be aimed at VE3DCU on top the bluff for satisfactory reception. The coaxial cable exits the waveguide from the rear; inside the waveguide is a small U-shaped fluorescent tube aligned with the waveguide center axis.

Engineering and Computer Department at McMaster University, Hamilton, Ontario, Canada, to whom the author is most grateful.

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Robert Zimmerman obtained his Novice license in 1965 at the age of 13. Being very timid as a youth, the entire year passed without his making a QSO. Upon receiving his General license, his ham radio efforts began in earnest.

Zimmerman has spent his entire career as an RF engineer. In 1979 he joined the engineering staff at Arecibo Observatory, designing cryogenic low-noise preamplifiers for the telescope. Later, at the NASA Goddard Spaceflight Center, he was a design engineer on the Mars Observer spacecraft.

Zimmerman holds an extra-class license, NP4B, and a Canadian advanced-class license, VE3RKZ. He enjoys 40 meter PSK31 and SSB. Presently he lives in the hills overlooking Arecibo, Puerto Rico. He may be reached at: HC 3 Box 21829, Arecibo, Puerto Rico 00612.

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